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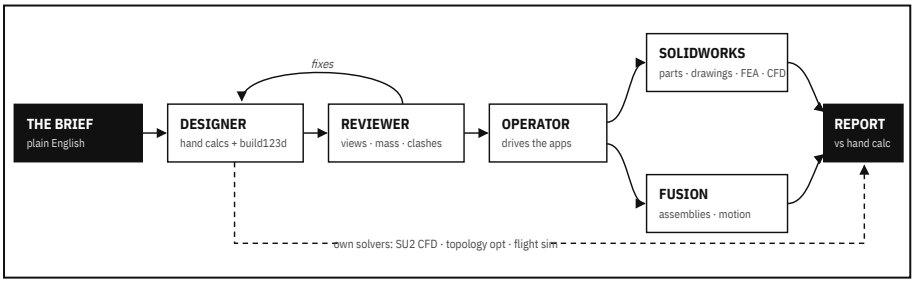
I built a pipeline that takes an engineering brief written in plain English and turns it into parametric CAD, SolidWorks drawings, stress, flow and thermal studies, and a report that checks every result against a hand calculation.

Sheet 1 is the tool. The later sheets are designs it produced from briefs I wrote, including work for my university Formula Student team and rocketry society. On those, the CAD, simulations and renders come from the pipeline. My part was the requirements, the review, and deciding what to fix. Each sheet says which parts were mine.

SHEET 1 CAD AND SIMULATION PIPELINE

Python · SolidWorks · Fusion · SU2

A set of AI agents with fixed jobs. A designer writes the part as Python code, a reviewer checks the views, mass and clashes, and an operator drives SolidWorks and Fusion through bridges into their APIs. Nothing goes in a report until the result has been compared with a hand calculation.



CAD AS CODE

Parts are build123d scripts with parameters, materials and self-checks. Every build exports STEP, a four-view sheet and a mass report.

SOLIDWORKS BRIDGE

20+ tools over the COM API: native features, BSI A3 drawings, assemblies with mates, and static, modal, thermal and Flow studies.

FUSION BRIDGE

An add-in that takes commands over localhost to import, colour, pose and screenshot assemblies. Its masses match SolidWorks exactly.

DATA

Built	**October 2026**
Code	**10,000+ lines of Python**
CAD	**build123d, SolidWorks 2025/26, Fusion**
Solvers	**SW Simulation, SW Flow, SU2, own FEM**
Benchmarks passed	**9 of 9**
Designs run through it	**7**

MY PART

- Set out what it had to do: brief in, checked design out, in the desktop CAD packages engineers actually use.
- Built it with Claude Code, directing the agents and testing each stage before moving on.
- Required a textbook benchmark for every solver before it touched a real design.

BENCHMARKS: EVERY SOLVER PATH AGAINST A KNOWN ANSWER

TEST	SOLVER	RESULT	REFERENCE	ERROR
Steel cantilever, tip load	SW static	deflection · stress	beam theory	+0.1 % · −4 %
Cantilever, first mode	SW modal	815.2 Hz	816.5 Hz	−0.2 %
Bar tip temperature	SW thermal	84.92 °C	84.88 °C	+0.05 %
Convecting fin tip	SW thermal	68.74 °C	68.7 °C	+0.06 %
Heated sphere in 2 m/s air	SW Flow	ΔT 20.99 K	21.17 K (Whitaker)	−0.8 %
Crank-slider position	SW mates + motion	slider x	slider-crank theory	0.0005 mm
Plate mass	Fusion bridge	mass	SolidWorks	exact
Transonic diffuser shock (Sajben)	SU2 RANS-SST	shock x/H* 2.25	2.25 (Bogar 1983)	wall p within 3 %
Solenoid field, axisymmetric	own FEM	B on axis	analytic	−0.13 %

WHAT WENT WRONG, AND WHAT CHANGED BECAUSE OF IT

1 Loads applied four times

A tank came back at four times the expected stress. Face selection was returning duplicates, so every pressure was applied once per copy. After deduplication: 56 MPa against a 45 MPa hand calc. Every study now gets a hand calc.

2 An intake that only looked started

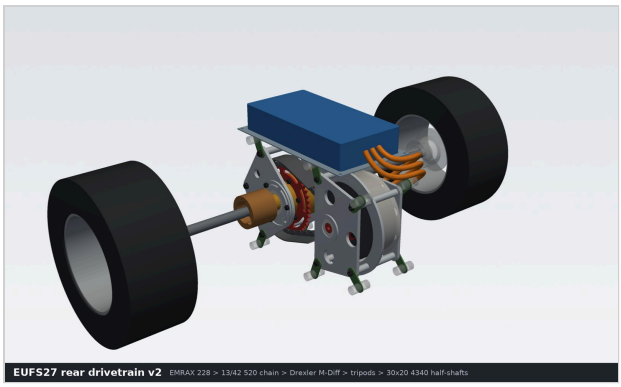
Off-design runs were first called started because they swallowed the design mass flow. Below Mach 3 the spillage hides that. The right test is the Mach number just ahead of the cowl lip, and it was subsonic, so the report was corrected.

3 Meshes that never finish

A knuckle mesh ran for 7.9 hours and failed. The pipeline now estimates element count before meshing and falls back to a simpler model when needed.

4 Questioning the brief

On the aerospike, the brief's mixture ratio and fuel flow disagreed, and no cooling layout could close the heat balance. Finding that on paper is the point.



2.1 Rear drivetrain, rev B. EMRAX 228 motor, 13:42 chain reduction, limited-slip diff on eccentric carriers, half-shafts to the wheels. 75 components, clash-free.

A chain-drive rear powertrain package for the university team's 2027 electric car: motor cradle, outrigger bearing, eccentric diff carriers, sprocket and half-shafts, built as a full SolidWorks assembly.

DATA

Reduction	**13:42 (3.23)**
Diff torque, design	**1,066 N·m (×1.5 shock)**
Chain pull, design	**9.95 kN**
Half-shaft	**30 × 20 mm 4340 tube**
Half-shaft FoS	**1.98 static · 1.71 fatigue**
Diff plate FoS on yield	**7.5**

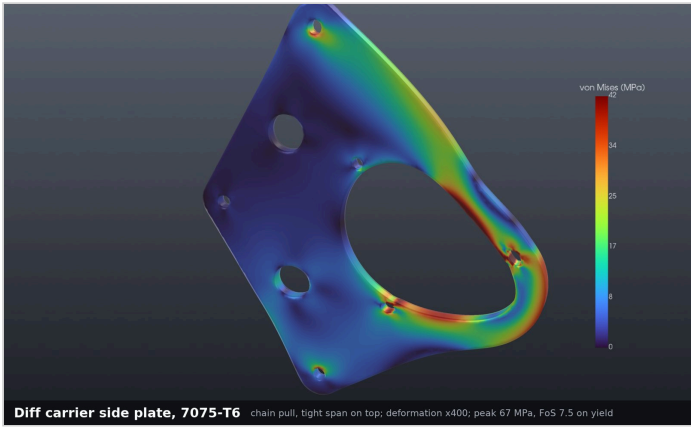
> Every mount passes. The 42-tooth sprocket's window corners set the limit, at a yield FoS of 1.32 after rev B, so the next step is a thicker or solid web. This is a concept for my subteam, not the final car.

MY PART

- Mechanical powertrain engineer on Edinburgh University Formula Student (EUFS).
- Set the brief: the most developed drivetrain package possible that fits the FS rules and gives the chassis team sensible hardpoints.
- Supplied the team's decisions on the bought-in motor, diff and inverter.

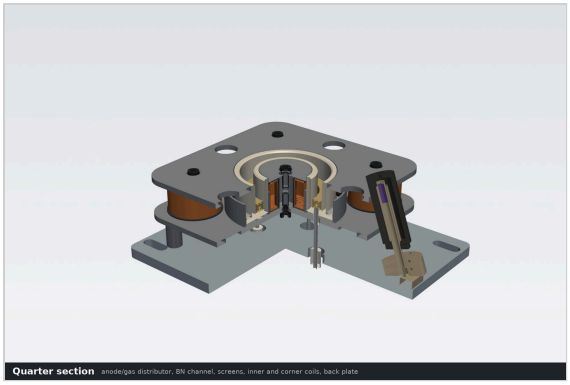
PRODUCED BY THE PIPELINE

- Motor cradle, outrigger, diff carriers, sprocket and half-shaft CAD, assembled natively in SolidWorks 2026.
- Hand calcs for shafts, splines, bolts and bearings; CalculiX FEA of the plates and sprocket.



2.2 Diff carrier side plate under chain pull (CalculiX, 7075-T6). Peak 67 MPa, FoS 7.5 on yield; deformation ×400.

HALL-EFFECT THRUSTER



3.1 Quarter section. BN discharge channel, stainless anode and gas distributor, Armco iron poles and coils, cathode tilted 20°.

A Hall thruster for Project Xenith, the Edinburgh University rocketry society's plan to fire a student-built thruster at Race 2 Space. Boron nitride channel, iron magnetic circuit with five coils, heaterless hollow cathode.

DATA

Power	**700 W**
Propellant	**Krypton**
Channel	**OD 68 / ID 48 mm**
Peak radial field	**200 G**
Field model vs analytic	**−0.13 %**
Anode temperature, est.	**~665 °C**

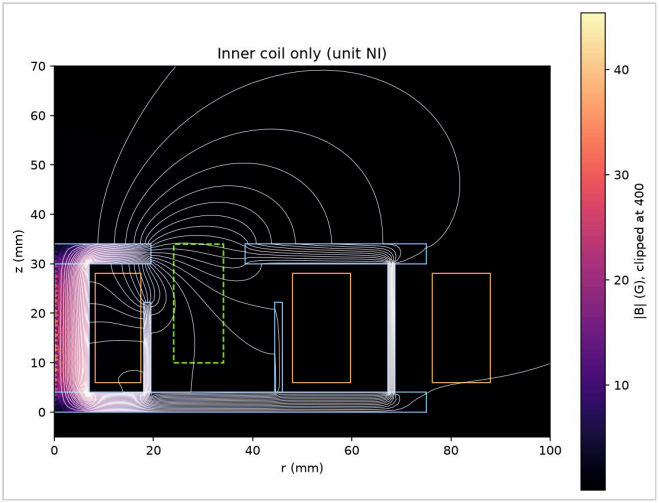
> The thermal network puts the inner coil at about 400 °C, well past polyimide-insulated wire, so it needs ceramic-insulated wire. Hardware tests come next: cold flow, field mapping and the cathode.

MY PART

- Thermal-mechanical engineer on Project Xenith, a 13-student team in Endeavour.
- Set the brief: a detailed SolidWorks thruster model in the SPT-70 class, with magnetics and thermal checks.

PRODUCED BY THE PIPELINE

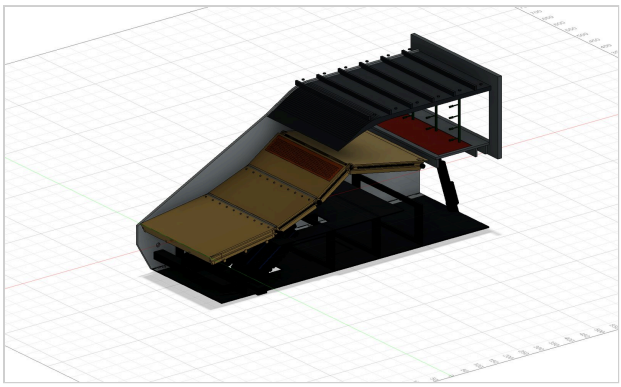
- Sizing from scaling laws, SolidWorks parts and assembly, bill of materials.
- Axisymmetric magnetics FEM checked against an analytic solenoid; radiation and conduction thermal network.
- Cold-flow test rig design.



3.2 Magnetic field, inner coil only. Own axisymmetric FEM: flux lines over field strength, used to place the 200 G peak just inside the exit.

SHEET 4 SUPERSONIC INTAKE

Mach 1.6 to 3.0



4.1 Mach 3 configuration, sidewall removed. Two variable ramps on ganged ballscrews, bleed plate behind the third ramp, aft ramp on a roller track.

An air intake that changes shape across its speed range so the engine always gets slowed, high-pressure air.

DATA

Capture area	**200 × 300 mm**
Ramps	**3 (2 variable)**
Pressure recovery, Mach 3	**0.833 (MIL spec 0.809)**
Actuator surge load	**12.9 kN**
Ramp 3 stress, p99	**136 MPa**

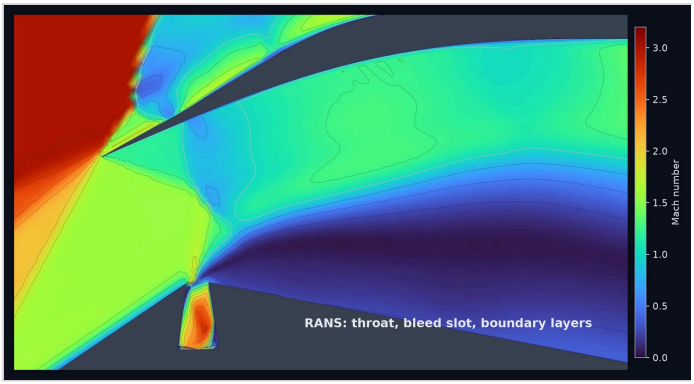
> Meets the spec at Mach 3. At Mach 2.0 and 2.4 it falls short because the cowl lip is fixed; a moving lip is the next step.

MY PART

- Wrote the brief: Mach 1.6–3.0, moving ramps on ballscrews, seals, bleed plate, bypass door, then CFD and thermo-mechanical FEA.
- Allowed changes to the brief where they helped, which led to three ramps.

PRODUCED BY THE PIPELINE

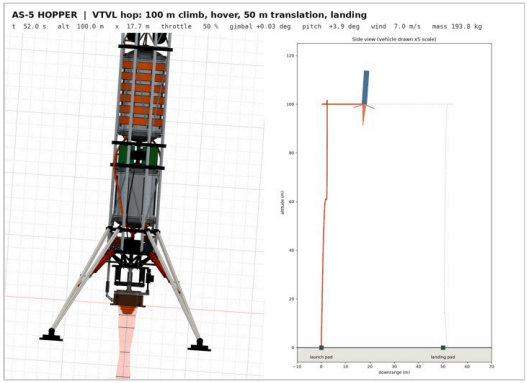
- Shock schedule, CAD and Fusion assembly, clash checks.
- SU2 CFD across the range, validated on the Sajben case first; ramp FEA and actuator hand calcs.



4.4 Throat, viscous model. The bleed slot runs as a supersonic jet; a separation bubble forms behind the throat.

SHEET 5 ROCKET HOP VEHICLE

VTVL, electric pump-fed



5.1 Simulated flight. Climb to 100 m, hover, move 50 m sideways through a gusting crosswind, land. Every number on screen comes from the flight-sim log.

A vertical take-off and landing test vehicle built around the aerospike on Sheet 6.

DATA

Take-off mass	**242 kg**
Thrust to weight	**1.62**
Hop apogee	**101.6 m**
Landing error	**2.0 m**
Parts in Fusion	**45**

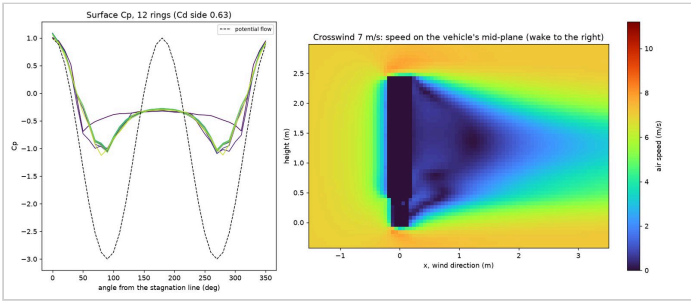
> Every structure passes with margin. The oxygen pump gets only 1.34× the suction head it needs, so it needs an inducer.

MY PART

- Set the brief: build a flying vehicle around the aerospike and gimbal, with a flight demo and full FEA and CFD.
- Chose the concept: a hop vehicle with electric pumps, designed in Fusion and checked in SolidWorks.

PRODUCED BY THE PIPELINE

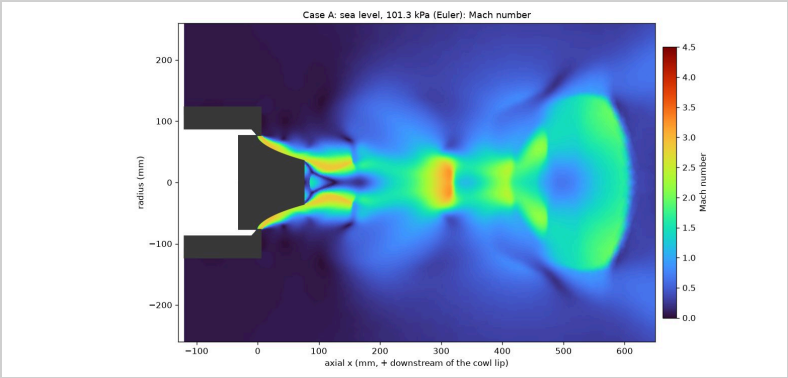
- Sizing, 45-part Fusion assembly, 3-DOF flight sim with attitude control.
- FEA of thrust structure, tank and legs; crosswind and feed-line CFD.



5.5 Crosswind at 7 m/s (SolidWorks Flow): 19.5 N·m of upsetting moment against 273 N·m of gimbal authority.

SHEET 6 AEROSPIKE ENGINE

5 kN, LOX / RP-1



6.1 Exhaust plume at sea level. The flow re-compresses onto the truncated spike, which is how an aerospike adapts to altitude.

An annular aerospike on a gimbal that steers it 6° in any direction.

DATA

Thrust, sea level	**3.83 kN**
Thrust, altitude	**5.0 kN**
Gimbal FoS	**3.4**
First mode	**66 Hz**

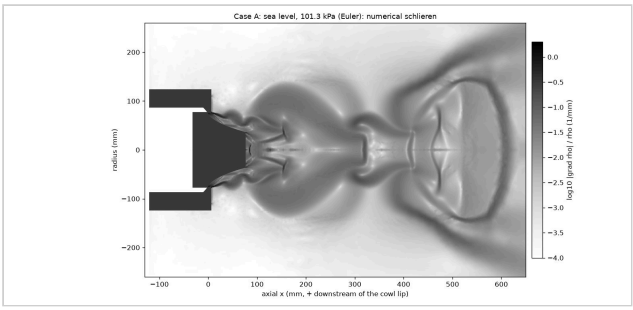
> Structurally sound, thermally not. Regen cooling can carry 0.46 MW against 0.9 MW in, so the cooling concept has to change.

MY PART

- Wrote the brief: a 5 kN annular aerospike with regenerative cooling and a gimballed mount, every relevant study in one overnight run.

PRODUCED BY THE PIPELINE

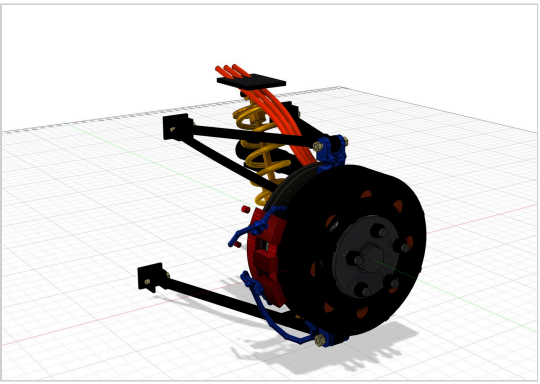
- Engine and two-ring gimbal CAD, kinematics, actuator sizing.
- Plume CFD checked against Angelino's method; gimbal FEA and modal; cooling optimiser.



6.4 Shock cells behind the spike base (simulated schlieren). Wall pressures match Angelino's design method.

SHEET 7 EV CORNER MODULE

7075-T6



7.1 Corner assembly. Wishbones, coilover, ventilated disc, caliper, hub, coolant and high-voltage lines.

A lightweight steering knuckle and in-wheel motor mount, shaped by topology optimisation.

DATA

Knuckle mass	**2.13 kg**
Bearing L10 life	**8,690 h**
Bump steer	**< 0.1°**
Bearing seat	**49–64 °C**

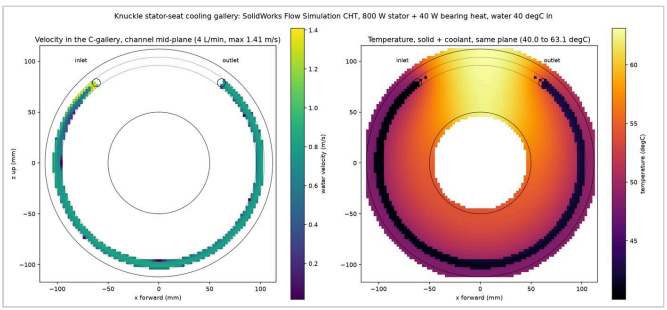
> Geometry and kinematics are done. The final stress check is paused after the mesher failed on the organic shape.

MY PART

- Wrote the brief: size a knuckle and motor mount from load cases, check bearing life and kinematics, optimise the shape, build the full corner.

PRODUCED BY THE PIPELINE

- 6-DOF suspension kinematics, a custom 3-D topology optimiser, the rebuilt knuckle.
- 33-part Fusion assembly and coolant-gallery heat transfer.



7.4 Motor-seat cooling (SolidWorks Flow, conjugate heat transfer): 800 W of stator heat keeps the seat between 40 and 63 °C.